

CSD 598 - Winter 2026

An example of exploration and inference

Pertti Palo

24 Feb 2026

First reflection I: Ethics

Any questions? Any thoughts? What did you find most difficult about this assignment?

First reflection II: Ethics

- ▶ Risk came up a lot unsurprisingly.
- ▶ I was impressed with the level of careful consideration you showed of different potential risk factors.
- ▶ You can also question the training itself. One of you brought up the absence of AI in the CORE materials.
- ▶ Contextualisation for the reader is good to do, but it's not absolutely necessary if the story of thinking is clear without the details. 'Keep the grader happy' is a good general rule in this context.

Last time: Visualisation to inferential statistics

Any questions? Any thoughts?

Exploration and inference example

- ▶ This comes from the latest data analysis I have run.
- ▶ As it is unpublished, I ask it not be recorded, and hence will not describe it here on the slides.
- ▶ I am happy to talk about it in detail and show how I've done the whole thing.

Model is a test, test is a model – sometimes

- ▶ ANOVA can be seen as a single predictor (controlled variable) linear regression model where the only predictor is a categorical variable.
- ▶ Simple regression – line fitting – can be seen as another way of testing for linear (Pearson's) correlation.

Model diagnosis – this will likely be moved to next time

- ▶ Collinearity (linear correlation) of the independent/predicting variables is bad (mathematically).
 - ▶ It can usually be spotted fairly easily by calculating correlations among the predictors and/or making scatter plots like a pairs plot.
- ▶ Heteroscedasticity will mess more the efficiency of prediction than the accuracy of it.
- ▶ Outliers can mess prediction accuracy.
- ▶ Here a bit of data exploration and model diagnostic plots go a long way:
 - ▶ Neat explanation of diagnostic plots can be found here
<https://library.virginia.edu/data/articles/diagnostic-plots>
 - ▶ Collinearity can be seen in for example a pairs plot or a correlation matrix (not to be confused with e.g. "Correlation of Fixed Effects" in lmer output).

Couple of nice resources

- ▶ A detailed look on ANOVA <https://statsandr.com/blog/anova-in-r/>.
- ▶ And ANOVA's cousin Kruskal-Wallis <https://statsandr.com/blog/kruskal-wallis-test-nonparametric-version-anova/>